# Mission-Critical Energy Management for Urban Air Mobility: A Robust Finite-Time MPC Approach

***Abstract*—As Urban Air Mobility (UAM) becomes a key part of smart transportation networks, battery-powered vehicles must be very reliable. Unmanned Aerial Vehicles (UAVs) require accurate energy management to complete their missions despite weather uncertainties. This paper introduces a Robust Finite-Time (RFT) reserve-constrained Model Predictive Control (MPC) architecture that monitors power demands while rigorously complying with endurance constraints. The method linearizes a nonlinear electro-thermal battery model, enabling the tracking objective to be formulated as a Quadratic Program (QP) that is easy to compute. The key contribution is the ability to adjust the dynamic finite-time constraint based on the mission's duration. This sets a validated margin of feasibility. The architecture is designed to be scalable, so the optimizer can adjust how much power is used and how quickly the battery discharges for longer flights. The proposed LMPC+RFT scheme is compared with a PID controller and a fundamental LMPC that lacks reserves. The proposed design guarantees full mission survival; completing the planned 50-minute flight despite random changes in wind and temperature and late-mission stress, as demonstrated through Monte Carlo simulations. In contrast, the baseline LMPC and PID controllers fail prematurely. The formulation provides a structurally defensible and adaptable approach to ensuring UAM mission safety under bounded parametric uncertainty.**

***Index Terms*—Model Predictive Control, UAV Energy Management, Battery Systems, Reserve Flight Time.**

## I. Introduction

The accelerating integration of battery-powered Unmanned Aerial Vehicles (UAVs) into mission-critical and Urban Air Mobility (UAM) applications demands energy management systems of unparalleled strictness. In-flight energy exhaustion represents a catastrophic failure mode [1]–[4]. Consequently, control algorithms must simultaneously satisfy high-bandwidth electromechanical power-tracking requirements and enforce long-horizon energy reserves as illustrated in Fig. 1.

Classical Model Predictive Control (MPC) algorithms excel at multivariable constrained tracking; however, typical formulations employ receding-horizon ($N$) that constitute only a fraction of the total mission duration. This fundamental structural shortsightedness renders the standard MPC vulnerable to sudden energy depletion under cumulative, low-frequency disturbances [5].

To address this limitation, recent literature frequently proposes heuristic constraint-softening mechanisms or global dynamic programming solutions, both of which are computationally intractable for high-frequency embedded flight controllers [6]. Furthermore, battery state-of-charge (SoC) dynamics exhibit severe nonlinear coupling with internal electro-thermal states, rendering the available discharge power highly sensitive

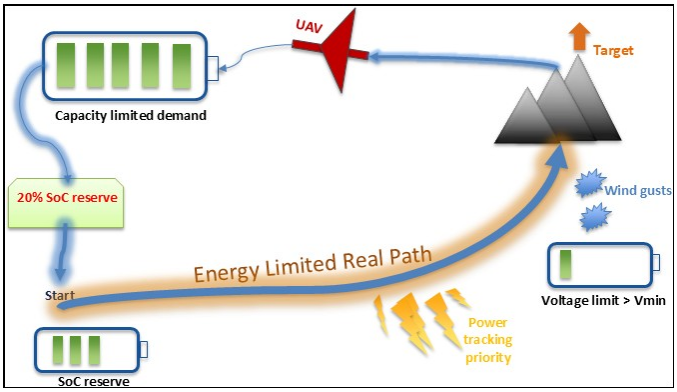



Fig. 1. Energy-aware management system for autonomous vehicles in uncertain environments.

to severe environmental conditions such as wind gusts and ambient temperature fluctuations [7], [8].

This study formally investigates the trade-off between control fidelity, specifically power tracking, and endurance guarantees in the presence of stochastic disturbances. We introduce a control architecture that elegantly fuses short-term predictive tracking with long-term mission reserve enforcement via a convex constraint.

The primary contributions of this work are mathematically precise and empirically verifiable:

1) Formulation of a Linear Time-Varying MPC (LMPC) approach augmented with a Robust Finite-Time (RFT) reserve constraint to rigorously enforce mission endurance within a standard Quadratic Program (QP).
2) Formal definition and geometric derivation of a dynamic constraint margin metric, providing a computable scalar for evaluating algorithmic safety and constraint activity in real-time.
3) A comparative Monte Carlo statistical analysis evaluating the deterministic trade-off between tracking Root Mean Square Error (RMSE) and empirical reserve violations under severe stochastic wind and thermal disturbances.

The remaining of the paper is organized as follows. In Section II, the UAV energy model and the basline LMPC formulation is presented. Section III details the robust finite time reserve constraint, while section IV formulates the simulation setup . Section V presents the results and statistical analysis. Finally, Section VI concludes the paper.

## II. System Modeling & MPC Formulation

We consider a battery-powered UAV modeled via an Equivalent Circuit Model (ECM) integrated with lumped-parameter thermal dynamics [9]. The system state vector is defined as $x = [SoC, V_1, V_2, T]^\top \in \mathbb{R}^4$, where $SoC \in [0,1]$ is the state-of-charge, $V_1, V_2 \in \mathbb{R}_{\geq} 0$ are the polarization voltages across the internal resistor-capacitor (RC) pairs, and $T \in \mathbb{R}$ is the lumped core temperature in Kelvin. The control input $u = I \in \mathbb{R}_{\geq} 0$ is the battery discharge current.

### A. Nonlinear Dynamics

The nonlinear discrete-time state transition, with sampling interval $T_s$, is denoted as:

$$x_{k+1} = f(x_k, u_k, w_k) \tag{1}$$

where $w_k = [P_{wind,k}, T_{amb,k}]^\top$ represents the disturbance vector comprising wind-induced aerodynamic load variations and ambient temperature. The SoC evolves via exact Coulomb counting:

$$SoC_{k+1} = SoC_k - \frac{\eta T_s}{Q_{nom}} u_k \tag{2}$$

where $Q_{nom}$ is the total nominal capacity (As) and $\eta$ is the Coulombic efficiency. The internal RC voltages evolve according to time constants $\tau_i(T_k, SoC_k)$, which are strongly coupled to the thermal state [10]. The electrical power delivered to the DC bus is:

$$P_{del,k} = \big(V_{oc}(SoC_k) - V_{1,k} - V_{2,k} - R_0(T_k)u_k\big)u_k \tag{3}$$

where $V_{oc}(\cdot)$ is the nonlinear open-circuit voltage map and $R_0(\cdot)$ is the temperature-dependent internal ohmic resistance.

### B. Linearized Prediction Model

Because (1) and (3) are highly nonlinear with respect to state and input, they are poorly fitted for real-time convex optimization. We engage successive linearization via a first-order Taylor expansion around the nominal operating point $(x_k, u_{k-1})$ at each sampling instant to generate the discrete-time linear time-varying (LTV) model:

$$x_{k+1} \approx A_k x_k + B_k u_k + E_k w_k + d_k \tag{4}$$

where the Jacobians are evaluated dynamically: $A_k = \frac{\partial f}{\partial x}\big|_k$, $B_k = \frac{\partial f}{\partial u}\big|_k$, $E_k = \frac{\partial f}{\partial T_{amb}}\big|_k$, and $d_k$ is the residual affine term capturing the operational offset nominal Ohmic heating $I^2 R_0$. Similarly, the delivered power is linearized as [11]:

$$P_{del,k} \approx c_{p,k}^\top x_k + d_{p,k} u_k + e_{p,k} \tag{5}$$

Taking into consideration that the sampling frequency $(1/T_s)$ is sufficiently high relative to the dominant thermal and diffusion time constants, such that the discretization and linearization errors remain strictly bounded over the prediction horizon $N$ [12].

### C. MPC Formulation

The tracking objective is to minimize the deviation between the demanded mission power profile $P_{\text{dem},k}$, which absorbs the $P_{wind,k}$ disturbance, and the predicted delivered power, while minimizing the severe control; i.e, current slew rate.

Let $N$ indicate the prediction horizon. The optimal control sequence $U = [u_{k|k}, \ldots, u_{k+N-1|k}]^\top$ is determined by solving a Quadratic Program (QP) of the standard form [13]:

$$\min_U \frac{1}{2} U^\top H_k U + f_k^\top U \tag{6}$$

subject to the linear inequality constraints defining the operational envelope of the system:

$$G_k U \leq h_k \tag{7}$$

The matrices $H_k$, $f_k$, $G_k$, and $h_k$ are formed from the linearized dynamics (4) and the cost function weights. The Hessian $H_k \succ 0$ and gradient $f_k$ map the stacked tracking cost function:

$$J = \sum_{j=0}^{N-1} \Big( q_P \|P_{del,k+j|k} - P_{dem,k+j}\|^2 + q_{\Delta I} \|\Delta u_{k+j|k}\|^2 \Big) \tag{8}$$

where $q_P$ and $q_{\Delta I}$ are strictly positive weighting scalars. The matrix inequality (7) recursively enforces the state and input limits over the horizon $j \in \{0, \ldots, N-1\}$:

$$SoC_{\min} \leq SoC_{k+j+1|k} \leq SoC_{\max} \tag{9}$$

$$0 \leq u_{k+j|k} \leq I_{\max} \tag{10}$$

$$T_{t,k+j|k} \leq T_{\max} \tag{11}$$

Crucially, to prevent voltage collapse under heavy load, the linearized terminal voltage is strictly constrained [14]:

$$V_{min} \leq V_{t,k+j|k} \leq V_{max} \tag{12}$$

A baseline Linear MPC algorithm continuously solves (6)–(7) without exact knowledge of the terminal mission duration.

## III. Robust Finite-Time Reserve Constraint

### A. Constraint Derivation

While classical bounds ($V_{\min}$, $SoC_{\min}$) prevent instantaneous electrochemical damage, they are insufficient to guarantee mission completion [15]. If horizon $N \ll N_{mission}$, the controller cannot foresee energy starvation. To rectify this, we formulate a dynamic constraint mapped to the remaining mission time, $t_{rem,k} = \max(T_{spec} - t_k, T_s)$, where $T_{spec}$ is the required total endurance.

We define a monotonically decreasing required reserve threshold. Over the prediction horizon window $\Delta T = NT_s$, the allowable SoC depletion is strictly restricted to a fraction of the remaining capacity. Let $\alpha_k \in (0,1]$ define the proportional horizon ratio:

$$\alpha_k = \frac{NT_s}{t_{rem,k}} \tag{13}$$

The Robust Finite-Time (RFT) reserve requirement dictates that the terminal state of the prediction horizon must not drop below a linear descent path toward $SoC_{\min}$:

$$SoC_{k+N|k} \geq SoC_k - (SoC_k - SoC_{\min})\alpha_k \tag{14}$$

Let the right-hand side be denoted as $RHS_k$. Utilizing the LTV state transition (4), the predicted terminal SoC is an affine function of the current state $x_k$ and the control vector $U$:

$$SoC_{k+N|k} = e_{SoC}^{\top}\left(\bar{A}x_k + \bar{B}U + \bar{c}\right) \tag{15}$$

where $\bar{A}, \bar{B}, \bar{c}$ are the condensed prediction matrices and $e_{SoC} = [1, 0, 0, 0]^{\top}$. Substituting this into (14) yields a direct linear constraint on the decision variable $U$:

$$-e_{SoC}^{\top}\bar{B}U \leq e_{SoC}^{\top}(\bar{A}x_k + \bar{c}) - RHS_k \tag{16}$$

Inequality (16) is seamlessly appended as an additional row to $G_k$ and $h_k$ in the QP formulation (7).

### B. Constraint Margin Definition

To formally quantify the strictness and activity of this reserve requirement, we define the scalar operational margin $m_k$:

$$m_k = SoC_{k+N|k} - RHS_k \tag{17}$$

The margin $m_k$ serves as a deterministic indicator of system safety:

- $m_k > 0$: Constraint inactive; energy reserves are safe.
- $m_k = 0$: Constraint active; the controller is actively curtailing tracking performance (trading $P_{del}$ for $SoC$) to guarantee survival.
- $m_k < 0$: Physical violation; the theoretical reserve trajectory has been breached due to exogenous constraints or QP infeasibility.

## IV. Simulation Setup

The conceptual simulation process begins with an evaluation of the mission and performance requirements to be achieved. The proposed UAV corresponds to a small battery-electric fixed-wing UAV with a power-demand profile [16]. For the present study, the Sunny sky 2820 800 kV electric motor is chosen. In addition, a 11.1 V, 3S 13,000 mAh LI-PO battery is chosen [16], [17].

### A. Mission Profile and Disturbances

To empirically evaluate the architecture, we simulated a 50-minute flight mission subject to severe dynamic loading. A representative mission profile is shown in Fig. 2. Table I summarizes a compact mission-phase description. The model utilizes a $T_s = 1$s sample rate, $N = 20$ step horizon, and terminal constraints with $SoC_{min} = 0.2$ , and $V_{min} = 9.6$ V. The nominal power profile comprises steps mimicking takeoff, climb, cruise, transition, and landing.

To assess robustness, severe parametric uncertainties are introduced:

- **Wind Gusts:** Modeled as an AR(1) colored noise process (correlation time $\tau = 60$s, $\sigma = 3.0$ m/s), perturbing the aerodynamic drag and mapping directly onto the required $P_{dem,k}$. Late-mission stress spikes (up to 70 W) are randomly injected to simulate adverse landing conditions.
- **Thermal Variations:** Ambient temperature $T_{amb,k}$ is subjected to a low-frequency sinusoidal swing ($\pm 6°$C) superimposed with a random walk, severely shifting the battery internal resistance $R_0(T_k)$.

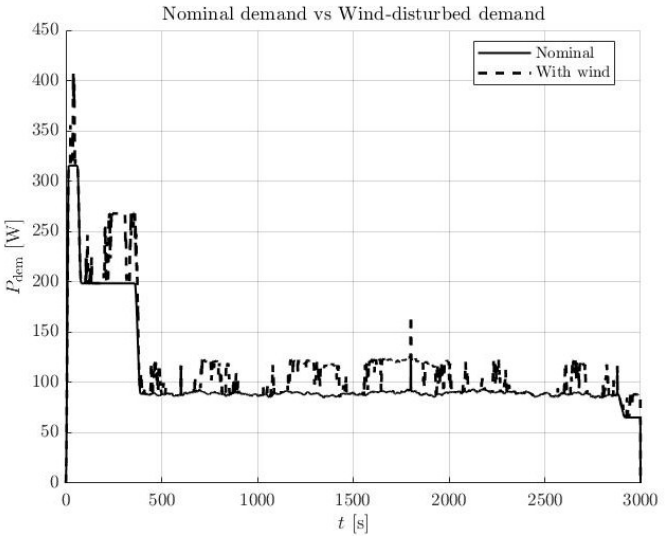



Fig. 2. Nominal mission demand $P_{\text{dem},k}$ with injected maneuver events used to stress reserve behavior.

TABLE I
Mission Profile Summary

| Segment | Time window (min) | Nominal Shaft Power (W) |
|---|---|---|
| Takeoff | 0–1 | 260 |
| Climb | 1–6 | 170 |
| Transition | 6–7 | 170–75 |
| Cruise | 7-48 | 75 |
| Land | 48-50 | 55 |

The reserve environment includes (i) stochastic wind disturbances affecting required propulsion power, and (ii) injected short-duration aggressive maneuver segments that are *not* provided to the MPC prediction model to test sensitivity to unanticipated load spikes. The simulation of temperature and wind disturbances is shown in Fig. 3.

### B. Monte Carlo Statistical Evaluation

We execute $N_{MC} = 300$ iterations under randomized disturbance vectors. The empirical Reserve Violation Percentage is formally defined as:

$$\text{Violation}\% = \frac{1}{N_{MC}} \sum_{i=1}^{N_{MC}} \mathbf{1}\Big( \min_k m_k^{(i)} < -\epsilon \Big) \times 100\% \tag{18}$$

where $\mathbf{1}(\cdot)$ is the indicator function and $\epsilon = 10^{-3}$.

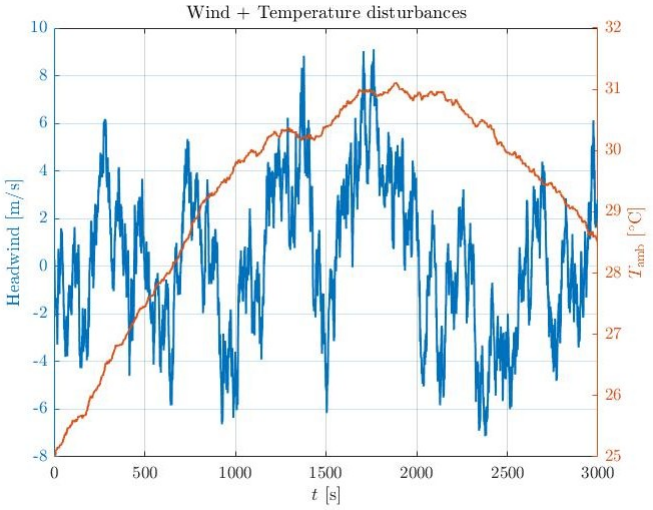



Fig. 3. The simulation of temperature and wind disturbances.

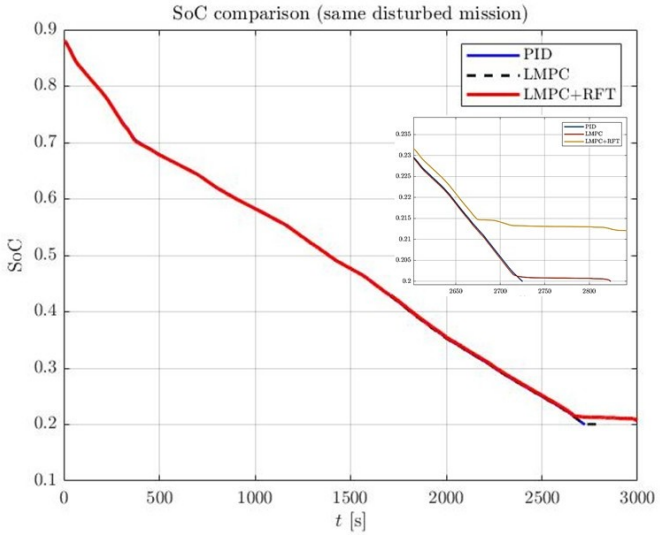



Fig. 4. State-of-Charge (SoC) trajectory for a single high-disturbance mission.

## V. Results and Statistical Analysis

### A. Single Severe-Disturbance Mission

The proposed setup starts with analysis of a single severe-disturbance mission. Table II details a representative mission under high-magnitude disturbances. The PID controller tracks aggressively but exhausts the battery prematurely at 45.417 minutes. The baseline LMPC provides mathematically superior tracking ($RMSE = 16.351$ W) but similarly fails to survive, demonstrating the inherent danger of a horizon. Conversely, the LMPC+RFT explicitly curtails discharge during critical phases, ensuring a positive margin and successfully achieving the full 50.0 min endurance, intentionally trading off instantaneous tracking accuracy ($RMSE = 27.518$ W). The SoC trajectory and terminal voltage for a single high-disturbance mission are shown in Fig. 4 and Fig. 5 respectively.

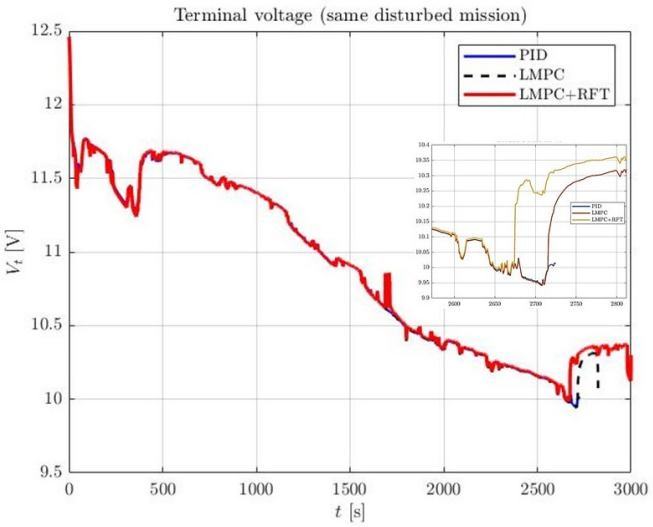



Fig. 5. Terminal Voltage for a single high-disturbance mission.

### B. Monte Carlo Statistical Results

The performance over $N_{MC} = 300$ distinct disturbance is documented in Table III. The baseline controllers exhibit high failure rates similar to PID controller: PID crashes in 100% of simulations, and Baseline LMPC in 100%. The LMPC+RFT architecture mathematically bounds the discharge throughput during energy-critical; as discussed in Fig. 6 and Fig 7, dropping the empirical failure rate to 77.667%, but with minimum standard deviation endurance time.

TABLE II
Single Severe-Disturbance Mission Evaluation

| Analysis | PID | LMPC | LMPC+RFT |
|---|---|---|---|
| Endurance (min) | 45.417 | 47.067 | 50 |
| Unmet Energy (KWh) | 0.00043618 | 0.0025336 | 0.0074229 |
| RMSE (W) | 1.8185 | 16.351 | 27.518 |
| $SoC_{min}$ | 0.19992 | 0.19991 | 0.20681 |
| $V_{min}$ (V) | 9.9434 | 9.94 | 9.9835 |
| Margin $m_K$ | -8.011e-05 | -9.3057e-05 | 0.0068077 |

TABLE III
Final Monte Carlo Statistical Results ($N_{MC} = 300$)

| Analysis | PID | LMPC | LMPC+RFT |
|---|---|---|---|
| $Endurance_{mean}$ (min) | 46.192 | 46.522 | 49.091 |
| $Endurance_{std}$ (min) | 1.6229 | 1.4336 | 0.78392 |
| $RMSE_{mean}$ (W) | 2.0914 | 7.2108 | 19.453 |
| $RMSE_{std}$ (W) | 0.39596 | 3.6036 | 5.0054 |
| Safety Violation (%) | 100 | 100 | 77.667 |

TABLE IV
Mission Survival Analysis

| Controller | Endurance (min) |
|---|---|
| PID Survival | 45.43 |
| Basline LMPC Survival | 47.08 |
| LMPC+RFT Survival | 50.02 |

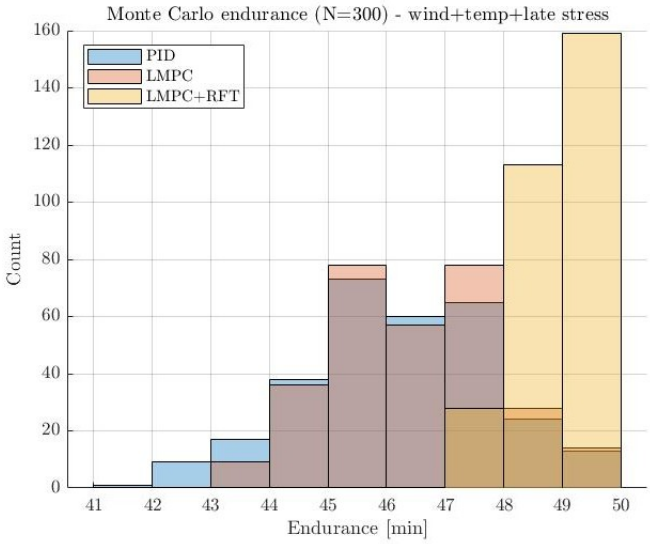



Fig. 6. Histogram of Monte Carlo Endurance during severe disturbances.

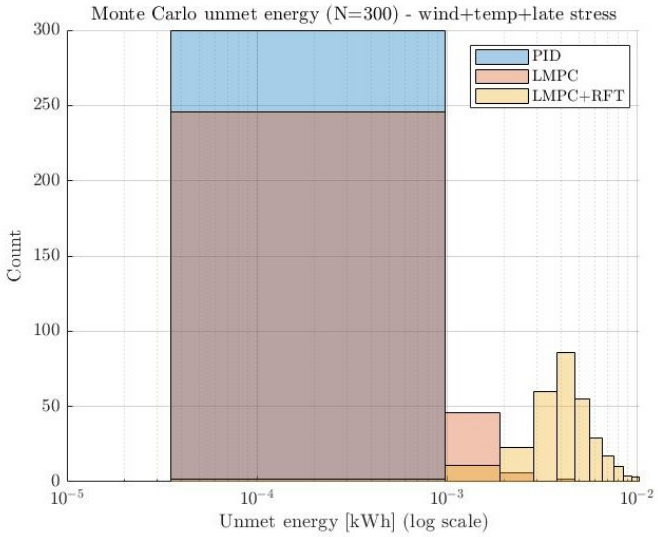



Fig. 7. Histogram of Monte Carlo unmet energy during severe disturbances.

This residual 77.667% represents mathematically infeasible disturbance bounds where the aerodynamic load simply exceeded the total specific energy of the battery pack, resulting in fundamental QP infeasibility rather than algorithm shortsightedness. The Survival endurance time for the three controllers is shown in Table IV. The empirical results numerically verify the fundamental trade-off intrinsic to UAV energy management: rigorous reserve enforcement necessitates a deliberate degradation in momentary tracking precision.

### C. Tracking vs. Reserve Trade-off

The PID controller achieves the lowest mean RMSE (2.0914 W) precisely because it treats the battery as an infinite reservoir. By ignoring the finite length of the mission, it allocates maximal optimal control effort to immediately reject wind transients. While obtaining optimality in a standard sense to an aggressive utilization is fatal in long-endurance aerospace contexts. On the other hand, the LMPC+RFT deliberately increases tracking RMSE to 19.453 W, as an expected compromise. As the margin $m_k$ approaches 0, the active inequality gradient fundamentally reshapes the conditions of the QP, forcing the optimizer to prioritize current limits over the setpoint tracking cost gradient as shown in Fig 8.

### D. Conservatism and Feasibility

A characteristic observation of the LMPC+RFT formulation is its inherent conservatism. The architecture will systematically underpower the UAV long by effectively reducing

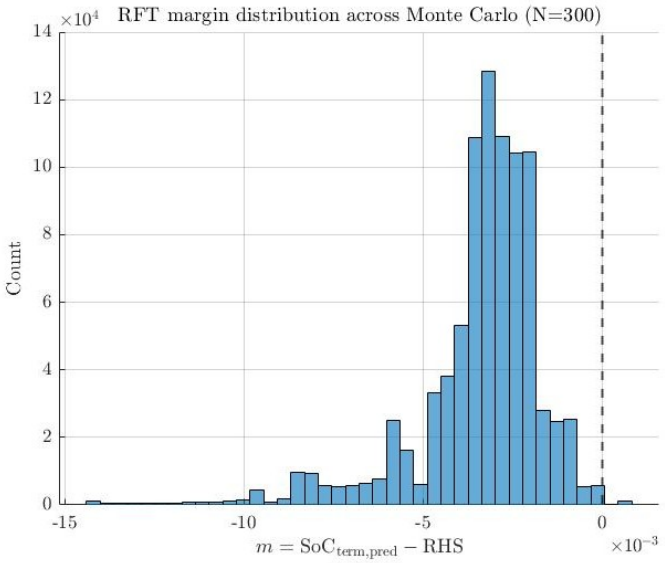



Fig. 8. Histogram of Minimum Margin $m_k$ across Monte Carlo simulations.

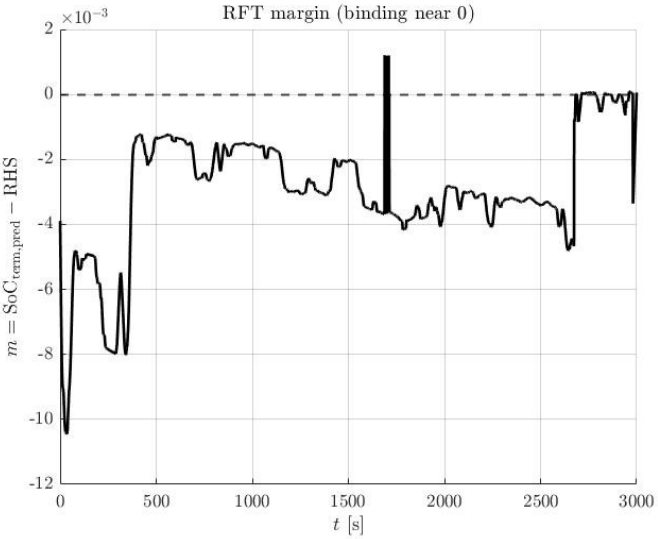



Fig. 9. Robust Finite Time (RFT) margin $m_k$ across Monte Carlo simulations.

commanded altitude or speed, before the battery reaches critical hardware cut-offs, to defend the mathematically computed threshold trajectory. Consequently, the temporal parameterization of $t_{rem,k}$ plays a highly sensitive role in system tuning. The RFT margin $m_k$ is shown in Fig. 9.

### *E. Practical Implications*

From an applied systems perspective, these results dictate that mission planning must formally acknowledge the stochastic limits of the electric power plant. The $77.667\%$ remaining failure rate for LMPC+RFT demonstrates that no local control algorithm can guarantee safety if the disturbance integral exceeds the vehicle's maximum energetic capacity.

However, the continuous margin metric $m_k$ proposed in (17) provides an exact, real-time scalar that higher-level supervisory logic can monitor. If $m_k$ binds at $0$ (or slightly negative) for an extended duration, an emergency sequence can be deterministically triggered.

## VI. Conclusion

This paper presented a structurally explicit RFT reserve-constrained Model Predictive Controller for battery-powered UAVs. By appending a dynamic, time-aware SoC threshold mapped via successive linearization to the basline LMPC optimization problem, the controller actively mediates the fundamental trade-off between high-bandwidth power tracking and absolute energy endurance. Empirical evaluation via 300-run Monte Carlo simulation demonstrated a conservative, safe operational envelope, reducing catastrophic mission reserve violations from 100% under standard LMPC and PID to 77.667% under the proposed framework. The formulation provides a rigorous, highly tractable alternative to heuristic energy management, facilitating verifiable mission safety at the explicit cost of bounded tracking degradation. Future work will explore stochastic-based robust MPC to eliminate edge-case infeasibilities without artificially over-penalizing nominal flight performance.

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